

Nibras Ahmed Nizar

MLOps Engineer

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Enterprise AI/ML Orchestration | Agentic AI & GenAI Systems | ML Pipeline & Deployment Ownership

SUMMARY

Senior MLOps Engineer and Technical Lead with over 6 years of experience delivering production AI/ML systems and automation pipelines for global semiconductor enterprises.

Proven track record owning the full ML lifecycle—from data ingestion and computer vision pipeline optimization to serverless model deployment and multi-agent GenAI applications.

Expert in architecting scalable solutions across hybrid on-prem and cloud environments.

SKILLS

Languages & Frameworks: Python, C++, SQL, TypeScript, Bash, FastAPI, React, Pandas, NumPy

ML & GenAI Orchestration: LangGraph, LangChain, MLflow 3.x, XGBoost, TensorFlow, PyTorch, RT-DETR, ONNX

MLOps & DevOps Infrastructure: Databricks (Model Serving, Asset Bundles, Unity Catalog), Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, OpenShift (ARO), AWS (SageMaker, EKS, Lambda, Glue, EventBridge, S3), Azure (ADLS Gen2)

Databases & Quality Assurance: PostgreSQL, SonarQube, Bandit, pip-audit, Pytest, OAuth M2M

EXPERIENCE

UST (Client: Applied Materials) • MLOps Engineer / Technical Lead

September 2025 - Present, Bengaluru

Led the design and rollout of production AI apps and MLOps infrastructure. Built multi-agent pipelines, moved models to serverless setups, and cut quarterly cloud spend by 50% while scaling to over 300 active users.

- **Agentic AI Ownership:** Designed and delivered "Defect Agent," a chat-first AI app linking 4 internal systems. Built a 3-node LangGraph pipeline with memory and WebSocket streaming on OpenShift/AKS and Databricks, growing the user base to 300+ active users with a 5,000+ target.
- **MLOps & Model Deployment:** Moved 6+ production ML models from fixed compute to serverless Databricks Model Serving. Used a custom MLflow PyFunc wrapper with scale-to-zero settings to cut idle compute costs and enable elastic scaling.
- **Data Pipeline Engineering:** Built an async, job-based replication pipeline using a 3-table staging model to safely move 7,000+ records and 21,000+ images into a central store. Automated hardware PDF ingestion into Databricks.
- **Dataset & Versioning:** Unified model and artifact versioning across the team using Unity Catalog volumes and MLflow 3.x, ensuring 100% lineage tracking from live models back to training runs.
- **Observability & Cost Control:** Set up production monitoring using MLflow tracing and inference tables to speed up debugging. Led a GPU optimization drive that cut quarterly cloud spend by 50% (from ~\$17K down to ~\$8.5K).
- **CI/CD & Security:** Created Jenkins and Terraform pipelines to enforce code quality gates (Pylint ≥ 9.0 , Bandit, 80%+ test coverage, SonarQube). Moved 15+ legacy notebooks to secure OAuth machine-to-machine authentication.
- **Technical Leadership:** Wrote the team's core MLOps standards and reusable LangGraph design patterns. Managed technical hiring by building an 8-part interview kit and running technical screens.

AI/ML Engineer (Freelance) — Saudi Arabia Client

September 2024 – May 2025, Remote

- **GenAI Lifecycle Pipeline:** Owned the end-to-end training pipeline for an Arabic GenAI assistant, fine-tuning LLMs via AWS SageMaker, tracking runs with MLflow, and serving models via FastAPI behind AWS API Gateway and Lambda.
- **Infrastructure & Automation:** Containerized application components on AWS EKS using GitHub Actions CI/CD; developed automated drift-triggered retraining jobs leveraging AWS Glue and EventBridge while achieving sub-100ms latency via ONNX.

INTEL • Software Engineer (AI/ML)

October 2021 - August 2024, Bengaluru

Led ML development and MLOps pipelines for Intel's hardware tools and computer vision teams. Built predictive models and automated workflows that cut design cycles by 15% and reduced model-integration latency by 20%.

- **MLOps & Pipeline Engineering:** Managed the MLOps stack for EDA toolchain ML components. Orchestrated Docker containers, MLflow tracking, and Jenkins CI/CD pipelines to drop model-integration latency by 20%.
- **Machine Learning Development:** Built and embedded the "TimeToSynthesize" XGBoost model into core FPGA tools, predicting hardware synthesis duration and cutting overall design-planning cycles by 15%.
- **Cross-Team Collaboration:** Partnered with the Intel Scenescape team to collect requirements and build end-to-end MLOps pipelines. Analyzed production computer vision models to optimize them for FPGA hardware deployment.
- **Verification & Performance:** Engineered a custom C++ formal-verification pipeline to validate logic across synthesis modes, cutting compile times by 30% to 50% for reusable IP blocks.
- **Leadership & Recognition:** Received two Intel "Above & Beyond" awards—Results Driven (2023) and Customer First (2022)—for cross-team delivery. Mentored 2 junior engineers.

PoleToWin India • Associate Engineer

March 2020 - February 2021, Bengaluru

- **ML in Quality Assurance:** Developed classification models (**Logistic Regression, Random Forest**) for defect prediction, enhancing detection accuracy by 15% and reducing test cycles by 20%.
- **Embedded Tooling Development:** Developed Python text-parsing and sensor-log analysis modules utilizing Pandas and NumPy, accelerating QA engineering debug lifecycles within an Agile cadence.

EDUCATION

Master of Technology (M.Tech) in Software Engineering

GPA: 9.2/10 | Graduated: 2022 | Ramaiah Institute of Technology, Bengaluru, India

Bachelor of Engineering (B.E.) in Computer Science & Engineering

GPA - 7.0/10 | Graduated: 2020 | Ramaiah Institute of Technology, Bengaluru, India

RELEVANT PROJECTS

- **Two-Stage Object Detection Pipeline (Coupon Counting):** Developed an automated stub-to-coupon detection pipeline using RT-DETR object detection deployed on Databricks with MLflow tracking, replacing an entirely manual auditing process.
- **Real-Time Telemetry Anomaly Detection:** Engineered a rolling-mean, EMA, autoencoder, and Robust Random Cut Forest ensemble model achieving 95% detection accuracy and 30% faster response times; secured a ₹1.8M ISRO development grant.
- **End-to-End Image Classification Pipeline (Personal):** Built an automated edge-AI MLOps pipeline using TensorFlow, FastAPI, Docker, and AWS, decreasing edge deployment timelines by 30% across 5+ simultaneous model versions.
- **Computer Vision-Based Object Detection (Personal):** Attained a 92% mAP metric utilizing YOLOv5 for real-time edge processing; optimized inference runtime beneath 100ms through ONNX compilation and mapped data onto a Streamlit ADAS visualization dashboard.

CERTIFICATIONS & AWARDS

- **Certifications:** AWS Certified Machine Learning – Specialty (2025) | MLOps Specialization, Duke University (2025) | Google AI Essentials (2025) | Introduction to Containers, Docker, Kubernetes & OpenShift, IBM (2023)
- **Awards:** Recipient of ₹1,800,000 ISRO "Respond Basket" Project Grant (2021–2022) | Intel "Above & Beyond" Institutional Winner (2022, 2023)